

Answers

1.0

2.3

3.D

4.D

Explanations

1.0

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

Also, the number of Young visitors = $2*2x = 4x$

$$\Rightarrow x+2x+4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

Let 'Y' be the number of Platinum tickets bought by the Young visitors.

Then, the number of Platinum tickets sold = $2Y$.

Consequently, we can say that the number of Gold tickets sold = $140 - 55 - 2Y = 85 - 2Y$.

	Platinum	Gold	Economy	Total
Old				20
Middle-aged				40
Young	Y		38	80
Total	2Y	85-2Y	55	140

Let us assume that 'Z' is the number of Economy tickets bought by the Old visitors. It is given that the number of Gold tickets bought by Old visitors was equal to the number of Economy tickets bought by Old visitors.

	Platinum	Gold	Economy	Total
Old	$20 - 2Z$	Z	Z	20
Middle-aged	$(Y+2Z) - 20$	$43 - (Y+Z)$	$17 - Z$	40
Young	Y	$42 - Y$	38	80
Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Gold tickets was strictly greater than the number of Young visitors buying Gold tickets.

$$Z > 42 - Y$$

$$\Rightarrow Z + Y > 42 \dots (1)$$

The number of Middle-aged visitors buying Gold tickets = $43 - (Y+Z)$

Since $(Y+Z) > 42$, then We can say that $(Y+Z)_{\min} = 43$.

Hence, the number of Middle-aged visitors buying Gold tickets = $43 - 43 = 0$

 VIDEO SOLUTION

2.3

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

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Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Economy tickets.

$$20 - 2Z = 17 - Z$$

$$\Rightarrow Z = 3$$

Therefore, we can say that the number of Old visitors buying Gold tickets = 3

 VIDEO SOLUTION

3. D

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = $2x$.

Also, the number of Young visitors = $2 \times 2x = 4x$

$$\Rightarrow x + 2x + 4x = 140$$

$$\Rightarrow x = 20$$

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Middle-aged	(Y+2Z) - 20	43 - (Y+Z)	17 - Z	40
Young	Y	42 - Y	38	80
Total	2Y	85-2Y	55	140

It is given that the number of Old visitors buying Platinum tickets was equal to the number of Middle-aged visitors buying Platinum tickets.

$$20 - 2Z = (Y+2Z) - 20$$

$$Y + 4Z = 40$$

$$2Y + 8Z = 80$$

$$2Y = 80 - 8Z$$

We can see that Z can take only integer values. Therefore, we can say that the the total number of Platinum tickets sold will be a multiple of 8. Hence, option D is the correct answer.

 VIDEO SOLUTION

4. D

Let 'x' be the the number of Old visitors. Then, the number of middle-aged visitors = 2x.

Also, the number of Young visitors = 2*2x = 4x

$$\Rightarrow x+2x+4x = 140$$

$$\Rightarrow x = 20$$

It is given that total of 55 Economy tickets were sold out.

It is given that Young visitors half the total number of Platinum tickets that were sold.

Let 'Y' be the number of Platinum tickets bought by the Young visitors.

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Young	Y	42 - Y	38	80
Total	2Y	85-2Y	55	140

Let us check with the help of options.

Option (A): The numbers of Gold and Platinum tickets bought by Young visitors were equal.

$$Y = 42 - Y$$

$$\Rightarrow Y = 21. \text{ Hence, this statement can be true.}$$

Option (B): The numbers of Middle-aged and Young visitors buying Gold tickets were equal

$$43 - (Y+Z) = 42 - Y$$

$\Rightarrow Z = 1$. Hence, this statement can be true.

Option (C): The numbers of Old and Middle-aged visitors buying Platinum tickets were equal

$$20 - 2Z = (Y + 2Z) - 20$$

$\Rightarrow Y + 4Z = 40$. Hence, this statement can be true.

Option (D): The numbers of Old and Middle-aged visitors buying Economy tickets were equal

$$Z = 17 - Z$$

$\Rightarrow Z = 8.5$. This is not possible as Z has to be an integer. Hence, we can say that this statement is false.

 VIDEO SOLUTION

